Title

A Knowledge Graph-Driven Digital Twin for Explainable Personalized Diabetes Risk Prediction

Project Description (300 words)

Diabetes is one of the leading chronic diseases worldwide, requiring early detection and continuous monitoring to reduce the risk of severe complications. Existing diabetes prediction systems are predominantly based on machine learning models that estimate disease risk from patient data. Although these models often achieve high predictive accuracy, they typically operate as black boxes, providing limited explanation of the factors contributing to the prediction and little support for personalized recommendations. This limits user trust and the adoption of AI-assisted healthcare.

This project aims to develop a **Knowledge Graph-driven Digital Twin prototype** for explainable personalized diabetes risk prediction. The prototype will utilize the **ShanghaiT2DM** datasets, which contains longitudinal clinical records of patients with Type 2 diabetes. A Digital Twin will create a virtual representation of each patient's health profile using clinical attributes such as age, body mass index, blood glucose level, blood pressure, lipid profile, insulin-related indicators, and other diabetes-related measurements. Machine learning algorithms, including Random Forest and XGBoost, will be evaluated to identify the best-performing model for diabetes risk prediction. A Personal Health Knowledge Graph will then be constructed using Neo4j to model relationships among patient characteristics, risk factors, diabetes conditions, treatment options, and evidence-based lifestyle recommendations. By integrating prediction results with the Knowledge Graph, the Digital Twin will provide interpretable explanations of diabetes risk and generate personalized recommendations.

The novelty of this project lies in integrating longitudinal patient modelling through a Digital Twin with semantic reasoning provided by a Personal Health Knowledge Graph, enabling the system to predict, explain, and personalize diabetes management. The project employs established technologies, including Python, Scikit-learn, Neo4j, and a lightweight web dashboard, making it achievable within two months. The prototype will be evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, and qualitative case studies to demonstrate the explainability and usefulness of the recommendations.

Research Track:

Exploratory / Early Research Track

Research Focus Selection:

Digital Health & Biomedical Applications

Stage 1: Objective > Prediction Only

Build: Patient Data → Machine Learning Model (Random Forest) → **Risk Prediction**

Output:

- Risk = Low / Medium / High, or
- Risk Probability = 84%

Stage 2: Objective > Prediction + Explanation

Build: Patient Data →Random Forest → **Risk Prediction** → **SHAP** → **Explanation**

Output:

- Current Risk = 84%
- Main factors: ↑ HbA1c ↑ BMI ↑ MeanCGM

Stage 3: Objective > Prediction + Explanation + Simulation (Digital Twin)

Build: Patient Data →Random Forest → **Risk Prediction** → **SHAP** → **Explanation** → **Digital Twin (Virtual Patient)** → **What-If Simulation**

Output:

- Current Risk = 84%
- Main factors: ↑ HbA1c ↑ BMI ↑ MeanCGM
- Simulated Interventions:
  - Scenario 1:
    - BMI: 32 → 27
    - Risk: 84% → 67%
  - Scenario 2:
    - HbA1c: 8.5 → 7.0
    - Risk: 84% → 49%
  - Scenario 3:
    - BMI: 32 → 27
    - HbA1c: 8.5 → 7.0
    - Risk: 84% → 35%

Key Contribution:
The system can estimate **how the patient's risk changes under different intervention scenarios**, enabling personalized "what-if" analysis.

Stage 4: Objective > Knowledge Graph-driven Digital Twin

Build: Patient Data →Random Forest → **Risk Prediction** → **SHAP Explanation** → **Knowledge Graph Reasoning** → **Intervention Generation** →**Digital Twin Simulation** → **Updated Risk Prediction**

Output:

- Current Risk = 84%
- Main factors: ↑ HbA1c ↑ BMI ↑ MeanCGM
- Knowledge Graph (KG) reasoning:
  - BMI = 32
    ↓
    Obesity
    ↓
    Insulin Resistance
    ↓
    Increase T2DM Risk
    ↓
    Recommend: ✓ Weight reduction ✓ Exercise ✓ Low carbohydrate diet
- KG-driven simulation:
  - Recommended Intervention:
    - Exercise + Diet Modification
  - Simulation:
    - BMI: 32 → 27
    - HbA1c: 8.5 → 7.0
  - Prediction Risk:
    - 84% → 35%

Key Contribution:

Unlike Stage 3 where intervention scenarios are manually defined, Stage 4 allows the **Knowledge Graph to automatically determine clinically meaningful interventions and drive the Digital Twin simulations.**

Recommended Datasets

- Pima Indians Diabetes Dataset
- Diabetes Health Indicators Dataset

Stage	Objective	Main Question Answered	Research Question (RQ)	Output	Main Contribution
Stage 1	Prediction Only	**What is the patient's current diabetes risk?**	**RQ1:** Can machine learning accurately predict diabetes risk from patient data?	Risk category (Low/Medium/High) or probability (84%)	Diabetes risk prediction model
Stage 2	Prediction + Explanation	**Why is the patient at high risk?**	**RQ2:** Which patient factors contribute most to diabetes risk prediction?	Risk score + feature importance (HbA1c, BMI, MeanCGM)	Explainable AI (XAI)
Stage 3	Prediction + Explanation + Simulation (Digital Twin)	**What will happen if the patient changes their health condition?**	**RQ3:** Can a Digital Twin simulate the impact of intervention scenarios on diabetes risk?	Updated risk after simulation scenarios	Personalized what-if simulation
Stage 4	Knowledge Graph-driven Digital Twin	**Which interventions should be applied and what are their effects?**	**RQ4:** Can a Knowledge Graph automatically identify personalized intervention strategies and drive Digital Twin simulations?	Recommended interventions + updated risk predictions	Knowledge-driven personalized decision support